

BEARTOOTH ISLAND, SK, Canada

WMO: 711340

Lat: 59.208N

Lon: 109.693W

Elev: 781

StdP: 14.29

Time Zone: -6.00 (W06)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.7	(c) -30.0	(d) -42.3	(e) 0.5	(f) -33.7	(g) -38.0	(h) 0.6	(i) -30.3	(j) 34.8	(k) -5.7	(l) 31.1	(m) -7.3	(n) 8.1	(o) 270	(p) 0.805

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.1	(c) 74.7	(d) 65.3	(e) 71.5	(f) 63.1	(g) 68.5	(h) 61.2	(i) 66.1	(j) 71.9	(k) 63.8	(l) 69.3	(m) 61.7	(n) 66.9	(o) 6.6	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	90.6	70.1	61.2	83.2	67.7	59.1	76.9	65.9	31.2	72.3	29.5	69.4	28.0	67.0	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.8	(b) 25.0	(c) 22.1	(e) -36.1	(f) 79.8	(g) 4.1	(h) 6.9	(i) -39.1	(j) 84.7	(k) -41.5	(l) 88.7	(m) -43.8	(n) 92.6	(o) -46.8	(p) 97.6		
(a) 28.8	(b) 25.0	(c) 22.1	(e) -37.0	(f) 68.2	(g) 4.3	(h) 3.2	(i) -40.1	(j) 70.5	(k) -42.6	(l) 72.4	(m) -45.0	(n) 74.2	(o) -48.1	(p) 76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	-9.2	0.2	10.4	26.7	38.6	50.8	60.5	59.6	49.0	35.9	18.8	0.5	
	DBStd	25.48	13.84	13.51	14.15	10.25	5.84	6.46	5.70	6.11	6.59	7.89	10.74	15.72	
	HDD50	8587	1834	1393	1229	698	357	64	2	6	95	439	936	1534	
	HDD65	13320	2299	1813	1694	1148	820	431	163	185	479	903	1386	1999	
	CDD50	786	0	0	0	0	2	87	327	304	66	0	0	0	
	CDD65	45	0	0	0	0	0	4	23	18	0	0	0	0	
	CDH74	117	0	0	0	0	0	12	58	46	1	0	0	0	
	CDH80	9	0	0	0	0	0	1	5	3	0	0	0	0	
Wind	WSAvg	10.8	10.6	10.6	10.6	10.8	11.0	9.9	8.6	8.9	11.3	13.1	12.3	11.8	
Precipitation	PrecAvg	17.9	0.9	0.8	1.0	0.9	1.3	2.2	2.7	2.5	2.2	1.4	1.0	0.9	
	PrecMax	22.2	1.9	1.9	2.7	2.5	2.5	3.4	4.6	5.0	5.7	2.5	2.1	2.0	
	PrecMin	13.8	0.4	0.2	0.1	0.0	0.5	1.4	1.5	0.9	0.8	0.4	0.4	0.4	
	PrecStd	2.6	0.4	0.4	0.5	0.5	0.5	0.6	0.8	0.9	0.9	0.6	0.5	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	35.3	40.6	48.4	61.9	75.2	78.7	78.6	69.2	53.0	38.3	33.4	
		MCWB	29.5	33.1	36.8	41.0	49.9	57.5	64.8	68.4	58.4	48.1	35.5	32.0	
	2%	DB	20.6	28.3	36.8	44.4	54.9	68.9	75.2	74.6	63.6	49.3	36.2	29.6	
		MCWB	19.8	26.7	34.1	39.1	46.7	57.0	65.4	66.4	57.1	45.9	34.7	28.8	
	5%	DB	14.1	23.5	33.3	41.6	50.1	64.4	72.2	71.2	60.2	47.0	34.3	24.9	
		MCWB	13.3	22.6	31.5	37.6	44.2	55.6	63.5	63.4	55.1	44.1	33.1	24.0	
	10%	DB	9.6	19.2	29.6	39.2	46.4	60.7	69.3	68.2	57.7	44.8	32.3	20.6	
		MCWB	9.0	18.6	28.0	36.1	41.7	53.3	62.1	61.5	53.3	42.3	31.3	19.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.1	33.2	37.3	41.5	51.7	60.8	68.4	69.7	59.5	48.7	35.9	31.8
			MCDB	31.1	35.4	39.9	46.9	58.0	70.1	74.6	75.4	66.4	52.0	36.9	32.8
2%		WB	20.8	26.4	34.5	39.3	47.1	57.6	65.8	66.9	57.1	46.3	34.8	27.7	
		MCDB	21.5	27.5	36.6	43.5	53.5	63.1	71.8	72.7	62.2	49.1	35.8	28.5	
5%		WB	14.0	21.7	31.8	37.6	44.0	55.5	64.0	64.5	55.0	44.4	32.9	23.0	
		MCDB	14.5	22.7	33.4	40.9	48.9	61.0	69.9	69.7	58.9	46.9	33.9	23.8	
10%		WB	9.2	16.5	28.3	36.0	41.6	53.1	62.1	62.4	53.4	42.8	30.6	18.9	
		MCDB	9.7	17.2	29.8	38.8	45.5	58.7	67.7	67.2	56.8	44.6	31.5	19.5	
Mean Daily Temperature Range	5% DB	MDBR	12.0	14.9	19.2	17.9	13.9	14.8	13.1	11.7	9.9	7.4	7.6	10.7	
		MCDBR	16.6	17.6	19.6	18.3	20.2	20.5	16.5	15.1	13.3	10.3	7.3	14.3	
		MCWBR	16.3	18.6	17.1	14.6	13.4	12.3	9.7	9.2	9.5	8.4	7.1	15.5	
	5% WB	MCDBR	17.5	19.5	19.1	17.7	18.5	18.2	15.2	14.2	12.2	10.3	7.8	15.0	
		MCWBR	16.9	18.6	17.3	14.1	13.1	11.9	9.8	9.4	9.0	8.3	7.2	14.7	
Clear-Sky Solar Irradiance	taub		0.207	0.237	0.267	0.314	0.332	0.361	0.385	0.346	0.312	0.278	0.222	0.183	
	taud		2.202	2.301	2.308	2.363	2.468	2.387	2.377	2.461	2.580	2.533	2.247	2.089	
	Ebn at Noon		207	259	281	284	285	277	266	270	266	243	196	171	
	Edh at Noon		13	20	28	32	31	34	34	29	22	17	12	10	
All-Sky Solar Radiation	RadAvg		131	387	897	1441	1761	1930	1744	1368	879	404	138	74	
	RadStd		17	34	57	97	157	136	111	77	82	44	19	7	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(a) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(a) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page